SOFTWARE ENGINEERING AND DESIGN PRINCIPLES (CS311L)

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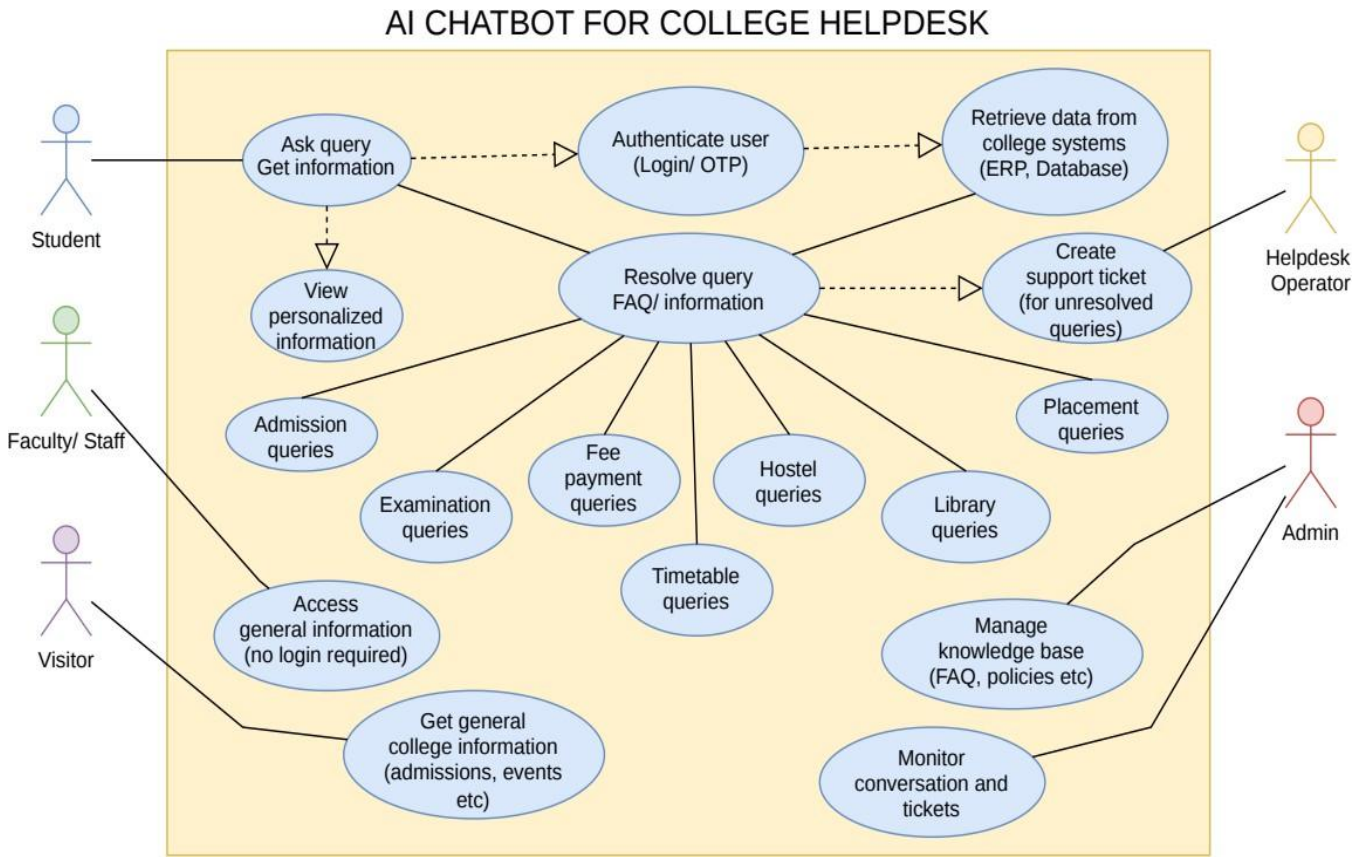
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AI CHATBOT FOR COLLEGE HELPDESK

Use Case Diagram, Actors and Use Case Specifications

1. Use Case Diagram

The following is the use-case diagram supplied for the AI Chatbot for College Helpdesk. It identifies Student, Faculty/Staff, Visitor, Helpdesk Operator and Admin as actors and shows their interaction with the chatbot system.



2. Actors and Their Roles

Actor	Role
Student	Uses the chatbot to ask general questions and, after authentication, access personalized information such as attendance, fee status and exam information.
Faculty/Staff	Uses the chatbot to obtain internal circulars, room-booking information and general administrative information.
Visitor	Uses the chatbot without login to access general college information such as admissions, events and contact information.
Helpdesk Operator	Handles escalated queries/tickets, monitors conversations, updates ticket status and helps resolve queries that the chatbot cannot answer.
Admin	Manages authorized administrative functions such as knowledge-base management, monitoring and system-level administration. The SRS distinguishes Helpdesk Admin/Operator and System Administrator roles.

3. Use Case Specifications

Each use case below contains its primary actor, function, precondition and postcondition. Where appropriate, the specification also notes the relevant system behaviour from the SRS.

Use Case	UC-01: Ask Query
Actor	Student, Faculty/Staff, Visitor
Function	Allows a user to enter a natural-language question. The system identifies the intent, extracts relevant entities and retrieves an appropriate answer or data.
Precondition	User has opened the chatbot and the chat interface is available. General queries do not require login.
Postcondition	The query is processed and a response is displayed. If the system cannot resolve it, clarification or escalation may be offered.

Use Case	UC-02: Get Information
Actor	Student, Faculty/Staff, Visitor
Function	Provides information about college services such as admissions, examinations, fees, timetable, hostel, library and placements.
Precondition	Chatbot is accessible and the user has submitted a valid query.
Postcondition	Relevant information is displayed to the user.

Use Case	UC-03: Access General Information
Actor	Visitor, Student (Guest)
Function	Allows users who are not logged in to access non-personal/general information such as admission procedures, general timetable and contact information.
Precondition	User can access the chatbot; authentication is not required.
Postcondition	General information is displayed and personal academic/fee information is not disclosed.

Use Case	**UC-04: Get General College Information**
Actor	Student, Faculty/Staff, Visitor
Function	Provides general college information, including admissions, events, examinations, hostel, library and placement information.
Precondition	User has opened the chatbot and the required information is available in the FAQ/knowledge base.
Postcondition	Requested college information is displayed; the user may provide helpful/not-helpful feedback.

Use Case	**UC-05: View Personalized Information**
Actor	Authenticated Student
Function	Allows a logged-in student to obtain personal information such as fee dues, attendance percentage and exam hall-ticket status.
Precondition	Student is successfully authenticated and the ERP/SIS API is available.
Postcondition	Student-specific information is retrieved and displayed.

Use Case	**UC-06: Authenticate User (Login/OTP)**
Actor	Student
Function	Verifies the student's identity when a query requires personal or academic information.
Precondition	Student has requested personalized information and is not currently authenticated.
Postcondition	Successful authentication permits access to personalized information; unsuccessful authentication denies such access.

Use Case	**UC-07: Retrieve Data from College Systems**
Actor	System / Authenticated Student
Function	Retrieves student-specific data from the college ERP/SIS, such as roll number, fee status, attendance and exam hall-ticket availability.
Precondition	Student is authenticated, valid student information is available and the ERP/SIS API is available.
Postcondition	Requested data is retrieved and presented. If the API is unavailable, an error is displayed and escalation may be offered.

Use Case	**UC-08: Resolve Query**
Actor	Chatbot / Helpdesk Operator
Function	Provides a solution automatically through the chatbot/knowledge base or manually through a helpdesk operator when escalation is required.
Precondition	A user query has been submitted and relevant information or a support process is available.
Postcondition	Query is answered/resolved, or a ticket is created and sent to a human operator.

Use Case	**UC-09: Create Support Ticket**
Actor	Student / Chatbot / Helpdesk System
Function	Creates a support ticket when the chatbot cannot resolve a query or the student explicitly requests a human agent.
Precondition	Query is unresolved, low-confidence, or the user selects 'Talk to a human'; required contact details and query summary are available.
Postcondition	A unique ticket is created, the student receives a ticket ID, and the ticket is routed to the appropriate helpdesk department.
Use Case	**UC-10: Manage Knowledge Base**
Actor	Helpdesk Operator / Admin
Function	Allows authorized staff to add, edit or delete FAQ and knowledge-base entries.
Precondition	Authorized user has authenticated dashboard access.
Postcondition	Knowledge-base information is updated and future chatbot responses can use the updated information.

Use Case	**UC-11: Monitor Conversations and Tickets**
Actor	Helpdesk Operator / Admin
Function	Allows authorized staff to monitor live/past conversations, view unresolved queries and track support tickets.
Precondition	Operator/Admin is authenticated and the dashboard is accessible.
Postcondition	Conversations and tickets can be viewed; ticket status can be updated and unresolved queries identified.

4. Additional Supporting Use Cases

Use Case	Actor	Precondition	Postcondition
Feedback on Answer	Student / Visitor	An FAQ answer has been displayed.	User submits helpful/not-helpful feedback; the response is logged.
View Analytics	Admin	Administrator is authenticated and has dashboard access.	Conversation and ticket statistics are displayed.
Update Ticket Status	Helpdesk Operator	A ticket exists and the operator is authenticated.	Ticket status is changed, such as Open, In Progress, Resolved or Closed.
Route Ticket	System / Helpdesk	A support ticket has been created and categorized.	Ticket is routed to the appropriate helpdesk department.
Access Chatbot	All users	Website/mobile chat interface is available.	Chat interface loads and can be used.
Clarify Query	Chatbot / User	Intent confidence is below the configured threshold.	The user provides clarification or the query proceeds toward escalation.

5. Actor–Use Case Mapping

Actor	Main Use Cases
Student	Ask Query; Get Information; Access General Information; Authenticate User; View Personalized Information; Create Support Ticket
Faculty/Staff	Ask Query; Get Information; Get General College Information; Create Support Ticket
Visitor	Ask Query; Access General Information; Get General College Information
Helpdesk Operator	Resolve Query; Monitor Conversations and Tickets; Manage Knowledge Base; Update Ticket Status
Admin / System Administrator	Manage Knowledge Base; Monitor Conversations and Tickets; View Analytics; Manage system/user roles